DocScout

Query-Aware Fine-Grained Evidence Retrieval from Long Documents

Project report: Information Retrieval course project

Team member	Roll number	Role
[Name 1]	[Roll 1]	Ingestion, parsing, indexing, data and qrels
[Name 2]	[Roll 2]	Retrieval, fusion, rerank, slots, table locator, graph context, evaluation
[Name 3]	[Roll 3]	LLM answering and grounding, feedback, API, React UI, tests

Every number in this report is copied from a file in the repository; the file is cited next to it. Nothing was re-run for this report and no LLM was called. Values that were never measured are marked **not recorded**.

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Member 2: [Name 2] ([Roll 2]): Retrieval, fusion, rerank, slots, table locator, graph context, evaluation . . . 21

Member 3: [Name 3] ([Roll 3]): LLM answering and grounding, feedback, API, React UI, tests . . . 21

1. Problem, objective and what DocScout does

Problem. Questions about long documents (reports, filings, forms) are usually answered at the level of a whole PDF or page. A reader still has to find the one paragraph, table cell or chart that holds the answer, and a generative answer without a pointer to that evidence cannot be checked.

Objective. Build a retrieval system that, for a question and a document, returns the exact evidence component (paragraph, table or table cells, figure) with provenance and a one-line "why retrieved" explanation, and a short answer that is grounded only in that evidence and cites it.

What DocScout does, in plain words. Each document is parsed once into components with page and box coordinates and indexed three ways (keyword BM25, dense BGE embeddings, CLIP image embeddings for figures). For a question it decides whether the question is textual, numerical or visual, searches every component of the document with all retrievers, fuses and reranks the results, finds the matching cells in tables, adds the surrounding context from the document's structure graph, and asks an LLM to answer only from the numbered evidence, citing it, or to say "Not found in the retrieved evidence". Every evidence item carries its page, bounding box, per-stage scores and graph path. A user can mark evidence relevant or not relevant; the list re-ranks with Rocchio feedback. A FastAPI service and a React UI show the answer next to the real page with the evidence highlighted.

Source: CLAUDE.md (project context); docs/reports/P6_REPORT.md, P7_REPORT.md

2. End-to-end flow

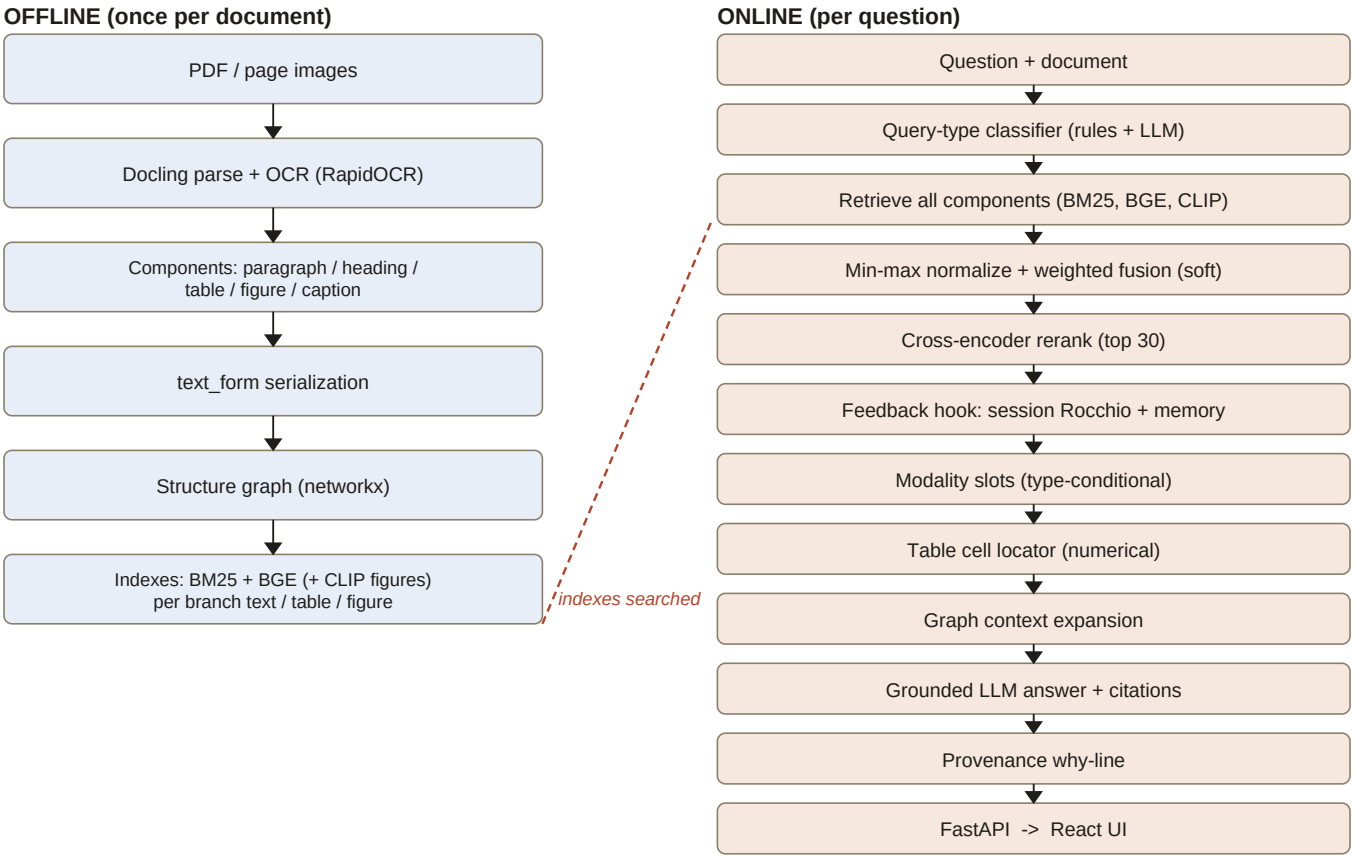


Figure 1. Offline indexing (left) and the online question path (right).

- **Ingest (Docling).** PDF pages (MP-DocVQA page PNGs are joined into one image-only PDF) are parsed by Docling with layout analysis, TableFormer and RapidOCR (onnxruntime backend). Items map to paragraph / heading / table / figure / caption; tables are saved to parquet and figures cropped to PNG (P2).
- **Components.** Each has doc_id, 1-based page, bbox in page pixels (top-left origin), page size, reading order, section and a *text_form*: raw text for paragraphs; title + caption + column headers + row headers + first 3 rows for tables; caption + nearest paragraph + inner text for figures (P2).
- **Structure graph.** Document -> Page -> Section (from headings) -> Block with parent-child, sibling and caption-of edges (P2).

- **Indexes.** Per branch (text = paragraph/heading/caption, table, figure): BM25 and BGE dense (FAISS); the figure branch adds CLIP image embeddings. MiniLM was built for comparison (P3).
- **Query-type classifier.** Word-boundary rules decide when exactly one group matches; otherwise the LLM decides (P5).
- **Hybrid weighted fusion (soft routing).** Every component of the document is scored by every retriever of its branch, min-max normalized per (branch, retriever) list and combined with retriever weights inside the branch and branch weights set by the query type. All branches are always searched (P5).
- **Cross-encoder rerank** of the fused top 30; the rest keep their fused order below (P5).
- **Feedback hook.** Identity by default; with votes it applies session Rocchio and (if on) the memory prior (P7).
- **Modality slots.** Type-conditional: a numerical question reserves a top-k slot for the best fused table, a visual one for the best fused figure (P6).
- **Table cell locator.** For numerical questions, rows and columns of the first tables in the evidence are matched to the question and the matching cells are returned (P6).
- **Graph context.** Section heading, caption or linked table/figure, and nearby reading-order siblings (P6).
- **Grounded answer.** The LLM sees only the numbered evidence, answers in JSON and must cite evidence ids; invalid ids are dropped and an answer without a valid citation becomes "not found" (P6).
- **Provenance.** Page, type, bbox, scores per stage, matched terms, graph path and a one-line why-line (P6).
- **Feedback.** Session Rocchio (re-score dense with q', re-fuse, rerank; up-votes pinned, down-votes removed) and an optional cross-session memory prior (P7).
- **FastAPI -> React UI.** /ask, /feedback, /next, /documents, /page-image; split-view UI (P7).

Source: docs/reports/P2_REPORT.md - P7_REPORT.md; src/docscout/pipeline.py docstring

3. Repository map

Every tracked file except generated evidence (docs/reports/p6_evidence, p7_evidence: saved command outputs and images) and eval/results (section 6). "Phase" is the first commit that added the file at its current path (git history; files moved in the post-P4 tidy-up show P4). Data folders are gitignored: data/raw (MP-DocVQA images and questions, demo PDF), data/processed (components, tables, figures, graphs, manifests, LLM cache, feedback.sqlite), data/indexes and data/indexes_demo.

Folder	Contents
src/docscout/	the Python package (config, schema, pipeline and one subpackage per stage)
src/docscout/dataset, ingestion, graph, indexing	offline: data download, parsing, serialization, graph, indexes
src/docscout/query, retrieval, rerank, localization, context	online retrieval stages
src/docscout/reasoning, evidence, feedback	LLM answering, provenance, relevance feedback
api/	FastAPI service
ui/	React + Vite + TypeScript front end
scripts/	thin command-line entry points (argparse + one call)
eval/	evaluation code; eval/results/ outputs
tests/	pytest suite (160 tests)
configs/	config.yaml
docs/	PHASES, PROGRESS, phase reports + evidence, final package

Path	What it contains	Phase	Lines
.claude/launch.json	dev-server entry for the UI preview (tooling)	P7	11
.env.example	template for LLM_PROVIDER / LLM_API_KEY / LLM_MODEL (real .env is gitignored)	P1	4
.gitignore	ignores data/, indexes, eval JSON outputs (except split.json), ui/node_modules, ui/dist	P1	28
CLAUDE.md	project brief, working rules and architecture	P1	143
README.md	setup on Windows, data, ingest, indexes, evaluation, API + UI, final deliverables	P1	179
api/__init__.py	package marker	P1	0
api/main.py	FastAPI: /ask, /feedback, /next, /documents, /page-image	P7	394
configs/config.yaml	every setting: paths, stores, models, fusion weights, classifier, localization, answer, feedback, llm, eval, api	P1	169
docs/PHASES.md	phase plan P1-P8 with acceptance criteria	P1	430
docs/PROGRESS.md	approved phase summaries	P1	261
docs/reports/P1_REPORT.md	phase report	P1	287
docs/reports/P2_REPORT.md	phase report	P2	365
docs/reports/P3_REPORT.md	phase report	P3	288
docs/reports/P4_REPORT.md	phase report	P4	287
docs/reports/P5_REPORT.md	phase report	P5	391
docs/reports/P6_REPORT.md	phase report	P6	600
docs/reports/P7_PREREGISTRATION.md	phase report	P7	25
docs/reports/P7_REPORT.md	phase report	P7	357
eval/__init__.py	package marker	P1	0
eval/build_qrels.py	page + component qrels	P4	180
eval/eval_answers.py	ANLS settings (a)/(b)/oracle, slot decision	P6	441
eval/eval_classifier.py	classifier accuracy / macro-F1	P5	220
eval/eval_feedback.py	simulated Rocchio feedback, pre-registered rule, memory test	P7	360
eval/eval_locator.py	table-locator cell hit rate	P6	197
eval/metrics.py	ranx wrapper, page-from-component, ANLS	P4	59
eval/plot_results.py	ablation.png	P5	84
eval/run_eval.py	8 retrieval systems, summary.md, t-tests	P4	444
eval/select_eval_docs.py	choose the 49-document evaluation subset	P4	80
eval/split.py	document-level DEV/TEST split	P5	77
eval/tune_fusion.py	DEV-only fusion weight grid	P5	125

Path	What it contains	Phase	Lines
pyproject.toml	package metadata (src layout, package docscout)	P1	19
requirements.lock.txt	pinned versions (CPU torch index on line 1)	P1	169
requirements.txt	Python dependencies	P1	42
scripts/__init__.py	package marker	P1	0
scripts/ask.py	entry: one question end to end (--demo, --k, --slots, --store, --json)	P6	91
scripts/build_indexes.py	entry: build indexes (--store, --rebuild)	P3	32
scripts/check_components.py	entry: validate components	P2	24
scripts/check_env.py	entry: library check	P1	12
scripts/download_data.py	entry: download MP-DocVQA subset	P1	28
scripts/explore_data.py	entry: dataset stats	P1	26
scripts/ingest.py	entry: parse + serialize + graph (--store, --limit)	P2	52
scripts/search_cli.py	entry: query one index	P3	33
scripts/visualize_components.py	entry: draw overlays	P2	36
src/docscout/__init__.py	package marker	P1	3
src/docscout/config.py	loads config.yaml, resolves paths, store switching (use_store), secrets from .env	P1	80
src/docscout/context/__init__.py	package marker	P1	0
src/docscout/context/graph_expander.py	section heading, caption/linked item, reading-order siblings	P6	104
src/docscout/dataset/__init__.py	package marker	P4	1
src/docscout/dataset/explore.py	dataset statistics	P4	88
src/docscout/dataset/mpdocvqa.py	download_subset(): MP-DocVQA val subset, page images, questions.jsonl	P4	262
src/docscout/env_check.py	library/version check (scripts.check_env)	P4	76
src/docscout/evidence/__init__.py	package marker	P1	0
src/docscout/evidence/cell_boxes.py	page boxes of localized cells (PyMuPDF, digital PDFs)	P7	83
src/docscout/evidence/provenance.py	per-stage scores, matched terms, graph path, why-line	P6	88
src/docscout/feedback/__init__.py	package marker	P1	0
src/docscout/feedback/memory.py	cross-session memory prior (one function)	P7	48
src/docscout/feedback/rocchio.py	session Rocchio FeedbackHook, pin / exclude, CE score cache	P7	182
src/docscout/feedback/session_store.py	SQLite votes table (evidence + answer votes)	P7	120
src/docscout/graph/__init__.py	package marker	P1	0
src/docscout/graph/builder.py	Document -> Page -> Section -> Block graph, caption links, helpers	P2	156
src/docscout/indexing/__init__.py	package marker	P1	0
src/docscout/indexing/base.py	BaseIndex interface, per-document rows, fingerprints	P3	104
src/docscout/indexing/bm25_index.py	tokenizer + BM25Okapi	P3	83
src/docscout/indexing/build_all.py	build/load all branch indexes, staleness check	P3	173
src/docscout/indexing/dense_index.py	sentence-transformers + FAISS IndexFlatIP (BGE, MiniLM)	P3	113
src/docscout/indexing/search.py	interactive search report (scripts.search_cli)	P4	57
src/docscout/indexing/visual_index.py	CLIP clip-ViT-B-32 index over figure crops	P3	64
src/docscout/ingestion/__init__.py	package marker	P1	0
src/docscout/ingestion/parser.py	Docling parse (layout, TableFormer, RapidOCR), type mapping, tables to parquet, figure crops, captions	P2	478
src/docscout/ingestion/runner.py	isolated ingest worker, crash/timeout recovery, cleanup, resumable run	P4	350
src/docscout/ingestion/serializer.py	text_form per component type (table/figure serialization)	P2	122
src/docscout/ingestion/validate.py	component checks (ids, bboxes, text)	P4	101
src/docscout/ingestion/visualize.py	debug overlays of component boxes	P4	77
src/docscout/localization/__init__.py	package marker	P1	0
src/docscout/localization/table_locator.py	row/column/cell selection in the top tables (numerical queries)	P6	288
src/docscout/pipeline.py	DocScoutPipeline.answer(): the 10 online steps, timings, meta.reranked	P6	253
src/docscout/query/__init__.py	package marker	P1	0
src/docscout/query/classifier.py	query type textual / numerical / visual: rules first, LLM when unsure	P5	120
src/docscout/reasoning/__init__.py	package marker	P1	0
src/docscout/reasoning/answerer.py	grounded answer, citation validation, not-found handling	P6	119

Path	What it contains	Phase	Lines
src/docscout/reasoning/llm_client.py	provider-agnostic LLM client (Groq), JSON mode, disk cache, retries, rate limits	P5	304
src/docscout/reasoning/prompts.py	numbered-evidence prompt, size cap	P6	152
src/docscout/rerank/__init__.py	package marker	P1	0
src/docscout/rerank/cross_encoder.py	ms-marco-MiniLM-L-6-v2 rerank of the top 30	P5	73
src/docscout/retrieval/__init__.py	package marker	P1	0
src/docscout/retrieval/fusion.py	weighted sum (main), RRF k=60, hard routing	P5	97
src/docscout/retrieval/normalize.py	min-max (default) and z-score per (branch, retriever)	P5	56
src/docscout/retrieval/retrievers.py	HybridRetriever: scores every component of the document; Candidate	P5	136
src/docscout/retrieval/slots.py	modality slot reservation (off / conditional / both)	P6	73
src/docscout/schema.py	Component / Evidence / QueryResult dataclasses, JSONL helpers	P1	156
tests/__init__.py	package marker	P1	0
tests/test_answerer.py	prompt cap, citations, not found, bootstrap (12 tests)	P6	105
tests/test_api.py	all endpoints (TestClient) (7 tests)	P7	143
tests/test_cell_boxes.py	cell boxes on a synthetic PDF (2 tests)	P7	36
tests/test_classifier.py	rules, LLM fallback (17 tests)	P5	74
tests/test_config.py	config paths, stores, keys (6 tests)	P1	68
tests/test_download_utils.py	download helpers (3 tests)	P1	29
tests/test_eval.py	qrels, metrics, page-from-component (11 tests)	P4	97
tests/test_graph_expander.py	context expansion (5 tests)	P6	63
tests/test_indexing.py	BM25 / dense / CLIP indexes, staleness (8 tests)	P3	137
tests/test_ingest_isolation.py	crashing worker is isolated (1 tests)	P4	37
tests/test_ingestion.py	parser mapping, serializer, graph (15 tests)	P2	229
tests/test_llm_client.py	cache, retries, rate limit (9 tests)	P5	123
tests/test_memory.py	votes store, memory gates (9 tests)	P7	101
tests/test_pipeline_smoke.py	end to end with fakes, feedback hook (6 tests)	P6	185
tests/test_retrieval.py	normalization, fusion, RRF, rerank tail (13 tests)	P5	169
tests/test_rocchio.py	Rocchio math, pin / exclude, lambda blend, identity (9 tests)	P7	129
tests/test_schema.py	Component / Evidence round trips, ids (5 tests)	P1	79
tests/test_slots.py	slot reservation variants (6 tests)	P6	77
tests/test_split.py	document-level split (2 tests)	P5	27
tests/test_table_locator.py	cell / row / column selection (14 tests)	P6	109
ui/__init__.py	package marker	P1	0
ui/index.html	Vite entry page	P7	12
ui/package-lock.json	pinned npm tree	P7	1628
ui/package.json	UI dependencies and scripts	P7	23
ui/src/App.tsx	layout, ask/vote/next flow, keyboard, FLIP re-rank animation	P7	264
ui/src/api.ts	typed API client	P7	95
ui/src/components/AnswerPanel.tsx	answer, citation chips, confidence strip, answer thumbs	P7	60
ui/src/components/EvidenceCard.tsx	evidence card: breadcrumb, cells, score bars, votes	P7	91
ui/src/components/PageViewer.tsx	page image with bbox and cell highlights	P7	72
ui/src/components/ScoreBars.tsx	bm25 / bge / clip / CE / fb bars	P7	54
ui/src/main.tsx	React root	P7	10
ui/src/styles.css	paper-and-ink theme, modality colours	P7	161
ui/tsconfig.json	TypeScript settings	P7	17
ui/vite.config.ts	dev server + /api proxy	P7	12

Source: git log --diff-filter=A; git ls-files

4. Each stage in detail

Stage	Input -> output	Key config values	Why this choice
Parse (Docling)	PDF -> components with bbox, tables (parquet), figure crops	parsing.ocr: true; ocr_backend: onnxruntime; pdf_backend mpdocvqa: pypdfium2, pdf: docling_parse; image_dpi 144; mpdocvqa_dpi 200	Layout + TableFormer give typed components with boxes. onnxruntime OCR: 27.4 -> 20.0 s/page (-27%), identical output (P2). pypdfium2 because docling_parse crashed natively on image-only PDFs (P4).
Serialize	component -> text_form	max_text_form_tokens 512; caption_max_gap 0.15	Tables/figures need searchable text: headers, row labels and caption carry most query terms (P2).
Graph	components -> networkx DiGraph	context.max_siblings 2; sibling_window 3	Gives section paths for provenance and context for the LLM (P2, P6).
Index	text_form / crops -> BM25, FAISS, CLIP	models.bge BAAI/bge-base-en-v1.5; minilm all-MiniLM-L6-v2; clip clip-ViT-B-32; bge_query_instruction on queries only	BM25 for exact terms and numbers, BGE for paraphrase, CLIP for charts (P3).
Classify	question -> textual / numerical / visual	classifier.use_llm true; rule_confidence 0.9; unsure_fallback textual	Rules alone 0.614 accuracy; rules + LLM 0.751 (P5, eval/results/classifier.json).
Retrieve + fuse	question -> every component scored	fusion.method weighted_sum; normalization minmax; retriever_weights bm25 0.7 / dense 0.3 / clip 0.3; branch_weights textual {0.4, 0.375, 0.225}, numerical {0.3, 0.6, 0.1}, visual {0.25, 0.15, 0.6}	Soft routing keeps all branches; weights tuned on DEV only (objective 0.4753 -> 0.5133, P5). Weighted beat RRF on component R@1 (0.432 vs 0.398 TEST).
Rerank	fused list -> top-30 reordered	models.cross_encoder ms-marco-MiniLM-L-6-v2; retrieval.rerank_top_n 30	Largest single gain: component MRR 0.599 -> 0.653 TEST (P5).
Slots	ranking -> top-k evidence	answer.slots conditional	Adopted by a pre-registered non-inferiority rule on DEV; no measurable benefit (P6).
Locate cells	top tables -> matching cells	localization.fuzzy_threshold 85; min_fuzzy_len 4; max_cells 12; top_k_tables 3	Numerical answers live in one cell; showing the cell is the finest evidence (P6).
Answer	numbered evidence -> JSON answer + citations	llm.model openai/gpt-oss-20b; temperature 0; max_tokens 512; answer.max_prompt_tokens 2500; not_found text	Grounding + citation validation; prompt cap keeps under Groq 8,000 TPM (P5, P6).
Provenance	candidate -> scores, path, why-line	-	Explains every item (P6).
Feedback	votes -> re-ranked list	feedback.rocchio alpha 1.0, beta 0.75, gamma 0.15, lambda 0.3; memory off; step 0.05; cap 0.15; min_votes 2; min_similarity 0.75; half_life_days 30	Rocchio is the classic relevance-feedback method; lambda term lets it reorder the CE top 30 (P7).
API + UI	HTTP -> JSON / page PNG	api.host 127.0.0.1; port 8000; cors_origins :5173	Models loaded once; UI shows page + evidence (P7).

Source: configs/config.yaml; P2-P7 reports; eval/results/classifier.json, tuning.json, summary.md

5. Datasets

5.1 MP-DocVQA subset

- Source: Hugging Face **lmms-lab/MP-DocVQA**, val split (fallback rubentito/mp-docvqa); the first 150 documents in stream order were downloaded (configs/config.yaml dataset; P1_REPORT).
- Evaluation subset (eval/select_eval_docs.py): documents with more than 20 pages skipped, stopping at about 250 pages: **49 documents, 250 pages, 293 questions** (246 on documents with >= 2 pages), 12 single-page documents (PROGRESS.md P4).
- Ingested: **2,976 components**: 2,380 paragraphs, 339 headings, 96 tables, 120 figures, 41 captions (PROGRESS.md P4).
- DEV/TEST split by document, seed 42, 20% of documents to DEV: DEV 10 documents / 76 questions; TEST 39 documents / 217 questions (176 on documents with >= 2 pages, 209 with a relevant component) (eval/results/split.json, PROGRESS.md P5). DEV is used for every tuning decision; TEST is looked at once per decision.
- Question types for analysis: gold labels from the DocVQA question_types field, joined on questionId (100% join, P5).

5.2 Demo PDF: Census P60-281

- "Health Insurance Coverage in the United States: 2022" (US Census Bureau, public domain): 31 pages, 1,892,540 bytes, text layer; Table 1, Figures 1-8, Tables A-1 to A-3 (P6_REPORT 4.1).
- Ingested into a separate **demo store** (stores.demo) so BM25 statistics of the evaluation corpus stay unchanged: 344 components (258 paragraphs, 58 headings, 9 tables, 13 figures, 6 captions), 281.8 s (P6_REPORT 4.2).
- Used for demos only; it has no gold answers and is not part of any metric.

5.3 How the qrels were built

- **Page relevance:** the question's answer_page.
- **Component relevance:** a component on the answer page whose text, text_form or a table cell contains a gold answer, after normalizing to [a-z0-9]. partial_ratio >= 90 only when the text is at least as long as the answer, else ratio; answers of <= 3 characters need an exact word or cell.
- **Coverage:** 278 of 293 questions (94.9%) have >= 1 relevant component; an audit removed a matching bug that had inflated it to 96.2%. Of 389 audited pairs, 298 were exact substrings, 77 exact short words/cells, 14 fuzzy OCR variants.
- Component metrics use those 278 questions (209 on TEST); page metrics use documents with >= 2 pages as the main number.

Source: PROGRESS.md P1/P4/P5; P4_REPORT.md; eval/results/split.json, qrels_info.json

6. Outputs explained

6.1 An evidence card from a real /ask -> /feedback session

Recorded in P7 (docs/reports/p7_evidence/api_session_top.json): the Medicaid question on p60-281 after the user down-voted the wrong table and up-voted Table 1. Fields, trimmed:


```
{
  "id": "p60-281_p8_table_0",
  "eid": "E1",
  "rank": 1,
  "type": "table",
  "page": 8,
  "branch": "table",
  "bbox": [104, 855, 1120, 1194],
  "page_size": [1224, 1584],
  "graph_path": "Page 8 -> Section 'Number and Percentage of People by He...' -> Table 0",
  "feedback_tag": "pinned",
  "scores": {"bm25_raw": 2.013, "bm25_norm": 0.411, "dense_bge_raw": 0.849, "dense_bge_norm": 1.0,
      "fused": 0.352, "rerank": 0.426, "feedback": 0.105, "final": 10.756},
  "localized_cells": [["Medicaid 2", "2022.Number", "62,050"], "... 9 more"],
  "cell_boxes": [{"row": "Medicaid 2", "column": "2022.Number", "value": "62,050", "bbox": [721,
      1129, 774, 1149]}]
}
```


why: numerical (table 0.6) | bm25 2.01->0.41, bge 0.85->1.00, fused 0.352, CE 0.43, fb +0.10 | pinned | terms: medicaid, coverage, 2022, thousands | Page 8 -> Section 'Number and Percentage of People by He...' -> Table 0 | bbox 104,855,1120,1194

Field	What it tells us
id / type / page / bbox / page_size	where the evidence is: component id, type, 1-based page, box in page pixels (top-left origin) and the page size the box refers to
branch	which index branch found it (text / table / figure)
graph_path	Page -> Section -> Block path in the structure graph (shown as a breadcrumb in the UI)
scores.bm25_raw / bm25_norm	BM25 score and its min-max value within the branch list (bar in the UI)
scores.dense_bge_raw / _norm	BGE cosine with the query and its normalized value
scores.fused	weighted-sum score: branch weight x (retriever-weighted normalized scores)
scores.rerank (CE)	cross-encoder logit for (question, text_form); decides the order of the top 30 when feedback is off
scores.feedback (fb)	change made by feedback on the normalized scale (Rocchio / memory), pin offset excluded
scores.final	the score the list is sorted by (raw CE when feedback is off; pinned items +10)
feedback_tag	rocchio / memory / pinned
localized_cells / localize_flag	(row label, column, value) cells chosen by the locator; flag cell / row / column / fallback / no_match
cell_boxes	page boxes of those cells (digital PDFs only); repeated values without a unique match get no box
why	the same information as one readable line: type and branch weight, raw->normalized scores, fused, CE, fb, tag, matched terms, path, bbox

6.2 The /ask response and the confidence strip

POST /ask returns query_id, session_id, query_type with the classification source and confidence, branch_weights, the answer object, the evidence cards, ranking_size (all components of the document), votes, memory state and feedback_active. answer.confidence drives the UI's confidence strip: **cited_sources** (number of valid citations), **not_found**, **slot_used** (a modality slot changed the evidence) and invalid_citations dropped. In the recorded session the first answer was "Not found" (0 cited sources, slot used) and after the two votes and Re-answer it was "62,050" with 1 cited source (P7_REPORT 4.6, 4.8).

6.3 eval/results/summary.md

One generated page with all headline tables: TEST page level on documents with >= 2 pages (main), TEST component level, TEST page level on all questions, paired t-tests vs dense BGE, the all-293 secondary tables, the ANLS table (P6) and the two relevance-feedback tables (P7). R@1 and MRR are the headline metrics; R@5/R@10 are near ceiling because documents are short (random pages already reach R@5 = 0.80 on TEST).

6.4 Each file in eval/results/

File	What it holds / what the numbers tell us
split.json	DEV/TEST document and question ids (the only committed result file)
qrels_page.json, qrels_component.json, qrels_info.json	relevance judgements in ranx format; per-question match details (which text matched which answer)
qtypes_gold.json, docvqa_qtypes_cache.json	gold question types from DocVQA question_types

File	What it holds / what the numbers tell us
baseline.json	random-page baseline per split and subset (expected R@k = mean(1/num_pages))
bm25_all.json, dense_minilm_all.json, dense_bge_all.json, hybrid_*.json	one retrieval system each: metrics per split (test/dev/all) x level (page_all, page_multi, component), latency median/p90, empty runs, full rankings
significance.json	ranx paired t-tests vs dense_bge_all
classifier.json	rules-only / LLM-only / combined accuracy and macro-F1, label counts, latency, predictions
tuning.json	DEV fusion grid (81 settings), objective before/after, chosen weights
ablation.png	bar chart of page/component R@1 and MRR per system
answers.json, answers_*.jsonl	ANLS per setting with bootstrap CIs, per-type scores, not-found rates; per-question answers
answers_slots_dev.json / _test.json, answers_*_slots-*.jsonl	slot-reservation comparison (DEV decision, TEST once)
locator.json, locator_dev*.json	table-locator cell hit rates in oracle and pipeline settings, miss categories
feedback_dev.json, feedback_test.json (+ _brief)	simulated Rocchio: residual metrics per setting, paired bootstrap CIs, rule outcome
feedback_memory_dev.json	memory test per ordered question pair (related / unrelated, fired, before/after metrics)

Source: eval/results/ (JSON outputs are gitignored and regenerated by the eval scripts; README 'Evaluation'); summary.md

7. Models

Role	Model / method	Compared against	Result (TEST unless noted)	Winner
Lexical retriever	BM25Okapi (rank-bm25), own tokenizer	BGE, MiniLM	component R@1 0.417 / MRR 0.552; page (>=2 p.) R@1 0.636 / MRR 0.718	best single retriever at component level
Dense embedder	BAAI/bge-base-en-v1.5 (query instruction)	all-MiniLM-L6-v2	page MRR 0.756 vs 0.725; comp MRR 0.544 vs 0.465; comp R@1 0.362 vs 0.259; median 35.6 vs 8.9 ms	BGE (+0.031 page MRR, +0.079 comp MRR)
Image retriever	CLIP clip-ViT-B-32 (figure crops)	not compared with another image model	qualitative only: 'bar chart' 3/3 correct on the demo PDF; scores nearly flat (0.22-0.25) (P6 4.4)	-
Fusion	weighted sum, min-max, soft routing	RRF (k=60) soft; weighted hard routing	comp R@1 0.432 vs RRF 0.398 vs hard 0.360; page MRR 0.788 vs 0.786 vs 0.696	weighted soft
Normalization	min-max	z-score implemented but **not compared**	-	-
Reranker	cross-encoder/ms-marco-MiniLM-L-6-v2 (top 30)	no rerank (no other reranker compared)	page MRR 0.788 -> 0.828 (p = 0.0002 vs BGE); comp MRR 0.599 -> 0.653; median latency 54 -> 763 ms	with rerank
Query classifier	rules + LLM (gpt-oss-20b) when unsure	rules only, LLM only	accuracy 0.614 / 0.747 / 0.751; macro-F1 0.521 / 0.644 / 0.652 (293 q)	rules + LLM
Routing input	predicted query type	gold DocVQA types (analysis)	comp MRR 0.599 predicted vs 0.607 gold (soft, no rerank)	gold slightly better (+0.008)
Answer LLM (evaluation)	openai/gpt-oss-20b via Groq, temp 0, JSON, reasoning_effort low	**no other LLM evaluated**; llama-3.1-8b-instant was requested but returns 404 on this key	ANLS (b) 0.646 [0.585, 0.707]	-
Answer LLM (demo only)	openai/gpt-oss-120b (scripts.ask --demo)	not evaluated	one demo run: same answer as 20b on the textual demo question (P6 4.5)	-
OCR backend	RapidOCR onnxruntime	RapidOCR torch	27.4 -> 20.0 s/page, identical text and boxes (P2)	onnxruntime
PDF backend (image-only)	pypdfium2	docling_parse	docling_parse crashed natively after 7 documents (P4)	pypdfium2
Feedback	Rocchio + lambda blend (beta 0.75, gamma 0.15, lambda 0.3)	11 other grid points; lambda = 0 literal; no feedback	residual MRR 0.312 vs 0.289 (+0.023 [-0.048, +0.091], not significant)	chosen by pre-registered rule on DEV

Not compared: other dense embedders (e.g. E5, GTE), other cross-encoders or LLM rerankers, other image models than CLIP ViT-B/32, other OCR engines or parsers than Docling + RapidOCR, BM25 parameters (k1, b), z-score normalization, and any LLM other than gpt-oss-20b on ANLS. gpt-oss-120b was never used in evaluation.

Source: eval/results/summary.md; classifier.json; P2, P4, P5 (2.1, 4.4-4.8), P6 (2.9, 4.4, 4.5), P7 reports

8. Evaluations by phase

8.1 P4 baselines (DEV + TEST: page level on documents with >= 2 pages, component level on 278 questions)

System	page R@1	page R@5	page MRR	comp R@1	comp R@10	ms/query
random	0.246	0.770	0.473	-	-	-
bm25_all	0.593	0.793	0.679	0.380	0.633	0.2
dense_minilm_all	0.537	0.902	0.696	0.244	0.729	9.4
dense_bge_all	0.606	0.907	0.738	0.331	0.726	35.4

Source: PROGRESS.md P4 (page level on documents with >= 2 pages; component level on 278 questions)

8.2 P5 retrieval systems (TEST)

System	page R@1	page MRR	page nDCG@10	comp R@1	comp MRR	comp nDCG@10	median / p90 ms
random	0.270	0.501	-	-	-	-	-
bm25_all	0.636	0.718	0.751	0.417	0.552	0.556	0.2 / 0.5
dense_minilm_all	0.568	0.725	0.788	0.259	0.465	0.524	8.9 / 11.8
dense_bge_all	0.631	0.756	0.810	0.362	0.544	0.582	35.6 / 40.4
hybrid_weighted_soft	0.659	0.788	0.839	0.432	0.599	0.624	54.3 / 60.2
hybrid_rrf_soft	0.659	0.786	0.837	0.398	0.581	0.612	54.8 / 60.5
hybrid_weighted_hard	0.602	0.696	0.740	0.360	0.473	0.458	36.1 / 43.3
hybrid_weighted_soft_rerank	**0.733**	**0.828**	0.869	**0.488**	**0.653**	0.677	762.7 / 1850.9
hybrid_weighted_soft_goldtype (analysis)	0.670	0.795	0.844	0.449	0.607	0.628	57.2 / 66.4

Source: eval/results/summary.md (TEST page level, documents with >= 2 pages; TEST component level)

Paired t-tests vs dense_bge_all: hybrid soft page MRR p = 0.0765 (not significant), component MRR p = 0.0032; soft + rerank page MRR p = 0.0002, component MRR p < 0.0001. Fusion weights were tuned on DEV only: objective (mean of DEV page MRR and component MRR) 0.4753 -> 0.5133 over 81 grid settings; the best point lies on the grid edge for BM25 = 0.7 and was not extended. Known limitation: with min-max a lone table gets 1.0 from both retrievers, so on DEV the hybrid page MRR (0.586) was below BGE (0.693); rerank mostly repairs it. Hard routing returns an empty ranking for 34/217 TEST questions.

Source: eval/results/summary.md; P5_REPORT 2.13, 4.4, 4.8; PROGRESS.md P5

8.3 P6 answer quality: ANLS (TEST, gpt-oss-20b)

Setting	n	ANLS [95% CI]	numerical	textual	visual	not found
(a) top-1 evidence only (baseline)	217	0.486 [0.424, 0.546]	0.515	0.466	0.344	38.2%
(b) top-3 + graph context + localized cells (final)	217	**0.646** [0.585, 0.707]	0.674	0.632	0.457	15.7%
(c) oracle: gold answer page as evidence	40	0.791 [0.666, 0.910]	0.778	0.798	1.000	15.0%

Paired (b) - (a) = +0.161 [+0.103, +0.218] on the same 217 questions; (b) better on 47, (a) better on 11. On the 40 oracle questions: (a) 0.545, (b) 0.740, oracle 0.791. 0 invalid citations in all settings. CIs: percentile bootstrap over questions, 2,000 resamples, seed 42.

Source: eval/results/summary.md (ANLS section); answers.json

8.4 P6 table cell locator (TEST, 62 numerical questions with a relevant table)

Version	oracle strict	oracle with row label	pipeline strict	pipeline with row label	DEV oracle strict
v1 base rules (clean estimate)	0.613 (38/62)	not measured	**0.548 (34/62)**	not measured	0.480 (12/25)
v3 shipped (two TEST-informed rules)	0.613 (38/62)	0.710 (44/62)	0.516 (32/62)	0.532	0.480 (12/25)

The whole table contains the answer in 95.2% of oracle cases; 13 of 24 oracle misses are answers that are row labels. In the pipeline 16 misses are questions not predicted numerical and 7 are wrong tables.

Source: P6_REPORT 4.6; eval/results/locator.json

8.5 P6 modality slots (pre-registered non-inferiority rule)

DEV (76 q): conditional - off = -0.005 [-0.015, +0.000] ANLS, locator unchanged (4/25) -> rule satisfied, type-conditional adopted with no measurable benefit. TEST once: 0.637 vs 0.646 (-0.009 [-0.023, +0.000]), 0 better / 2 worse. The demo check in P7 (k = 5) showed the reserved table/figure was the wrong one for both demo questions; no answer changed.

Source: P6_REPORT 4.10; P7_REPORT 4.1

8.6 P7 relevance feedback (simulated, residual collection, component level)

Pre-registered before any DEV run (docs/reports/P7_PREREGISTRATION.md): grid beta {0.5, 0.75} x gamma {0, 0.15, 0.25} x lambda {0.3, 0.5}; keep the default (0.75, 0.15, 0.3) unless another point gains >= 0.01 DEV residual MRR without losing R@1. Round 1 = final pipeline; the simulated user votes on the top 5 from the qrels; judged items are removed from both rankings (residual collection). The rule kept the default.

Split / system	n	R@1 (diff [95% CI])	MRR (diff [95% CI])	nDCG@5 (diff [95% CI])
DEV no feedback	34	0.059	0.202	0.226
DEV lambda = 0 (literal)	34	0.074 (+0.015 [+0.000, +0.044])	0.229 (+0.027 [-0.002, +0.083])	0.244 (+0.018 [+0.000, +0.054])
DEV chosen b0.75 g0.15 l0.3	34	0.191 (+0.132 [+0.029, +0.250])	0.293 (+0.091 [+0.011, +0.180])	0.270 (+0.044 [-0.027, +0.115])
TEST no feedback	68	0.120	0.289	0.271
TEST lambda = 0 (literal)	68	0.110 (-0.010 [-0.044, +0.015])	0.294 (+0.004 [-0.020, +0.027])	0.271 (+0.000 [-0.018, +0.018])
TEST chosen (run once)	68	0.140 (+0.020 [-0.069, +0.101])	**0.312 (+0.023 [-0.048, +0.091])**	0.295 (+0.024 [-0.040, +0.089])

The TEST gain is not significant (every CI includes 0); only 16 of 68 TEST residual questions had an up-vote in the top 5. Round 1 reproduces P5 (TEST component R@1 0.488, MRR 0.653), so feedback-off numbers are unchanged.

Source: eval/results/summary.md (feedback sections); P7_REPORT 4.2-4.4

8.7 P7 cross-session memory test (DEV, 598 ordered question pairs on 10 documents)

Group	pairs	fired	MRR diff [95% CI]	R@1 diff	better / worse
related (share a gold component)	172	22	+0.0070 [+0.0003, +0.0161]	+0.0116	5 / 0
related, where the prior fired	22	22	+0.0551 [+0.0021, +0.1220]	+0.0909	5 / 0
unrelated	426	34	+0.0003 [-0.0008, +0.0015]	+0.0000	4 / 5
unrelated, where the prior fired	34	34	+0.0033 [-0.0094, +0.0185]	+0.0000	4 / 5

Related questions improve where the gates fire; unrelated questions are flat on average but 5 of 34 pairs moved down below rank 1 (worst rank 3 -> 4). The 0.75 similarity gate passes only 22 of 172 related pairs (median same-document cosine 0.51).

Source: P7_REPORT 4.5; eval/results/feedback_memory_dev.json

8.8 Final headline results (TEST)

Metric	Value	Compared with	Source
Component R@1 / MRR (final pipeline)	**0.488 / 0.653**	dense BGE 0.362 / 0.544 (MRR p < 0.0001)	summary.md
Component nDCG@5 (final pipeline)	**0.644**	BGE: not recorded	P7_REPORT 4.3 (no-fb full ranking)
Component nDCG@10 (final pipeline)	0.677	BGE 0.582	summary.md
Page R@1 / MRR, documents >= 2 pages	**0.733 / 0.828**	BGE 0.631 / 0.756; random 0.270 / 0.501	summary.md
Page nDCG@5	not recorded	page nDCG@10 = 0.869	summary.md
ANLS: baseline (a) / final (b) / oracle	**0.486 / 0.646 / 0.791**	(b)-(a) +0.161 [+0.103, +0.218]	summary.md
ANLS with shipped default slots: conditional	0.637	slots off 0.646	P6_REPORT 4.10
Classifier accuracy (rules + LLM)	0.751	rules only 0.614	classifier.json
Table locator cell hit rate (pipeline, clean v1)	0.548 (34/62)	oracle 0.613	P6_REPORT 4.6
Rocchio residual MRR	0.312	no feedback 0.289; +0.023 [-0.048, +0.091], **not significant**	summary.md
Median latency, retrieval + rerank	762.7 ms (p90 1,850.9)	no rerank 54.3 ms	summary.md

9. UI walkthrough

Start the API and the UI (section 12), open http://localhost:5173, pick a document in the dropdown (the demo PDF is first, then the 49 evaluation documents), and ask a question or click a sample question. Left: the answer with citation chips and the confidence strip, then the evidence cards (rank, type, page, breadcrumb, localized cells, score bars, why-line on demand). Right: the real page image with every evidence box in its modality colour (text blue, table vermilion, figure green) and the cited cells highlighted. Vote with the buttons or the keyboard (J/K move, up/down vote, N next); the list re-ranks without a new LLM call, and "Re-answer with current evidence" asks again in the same session. The memory switch turns on the cross-session prior.

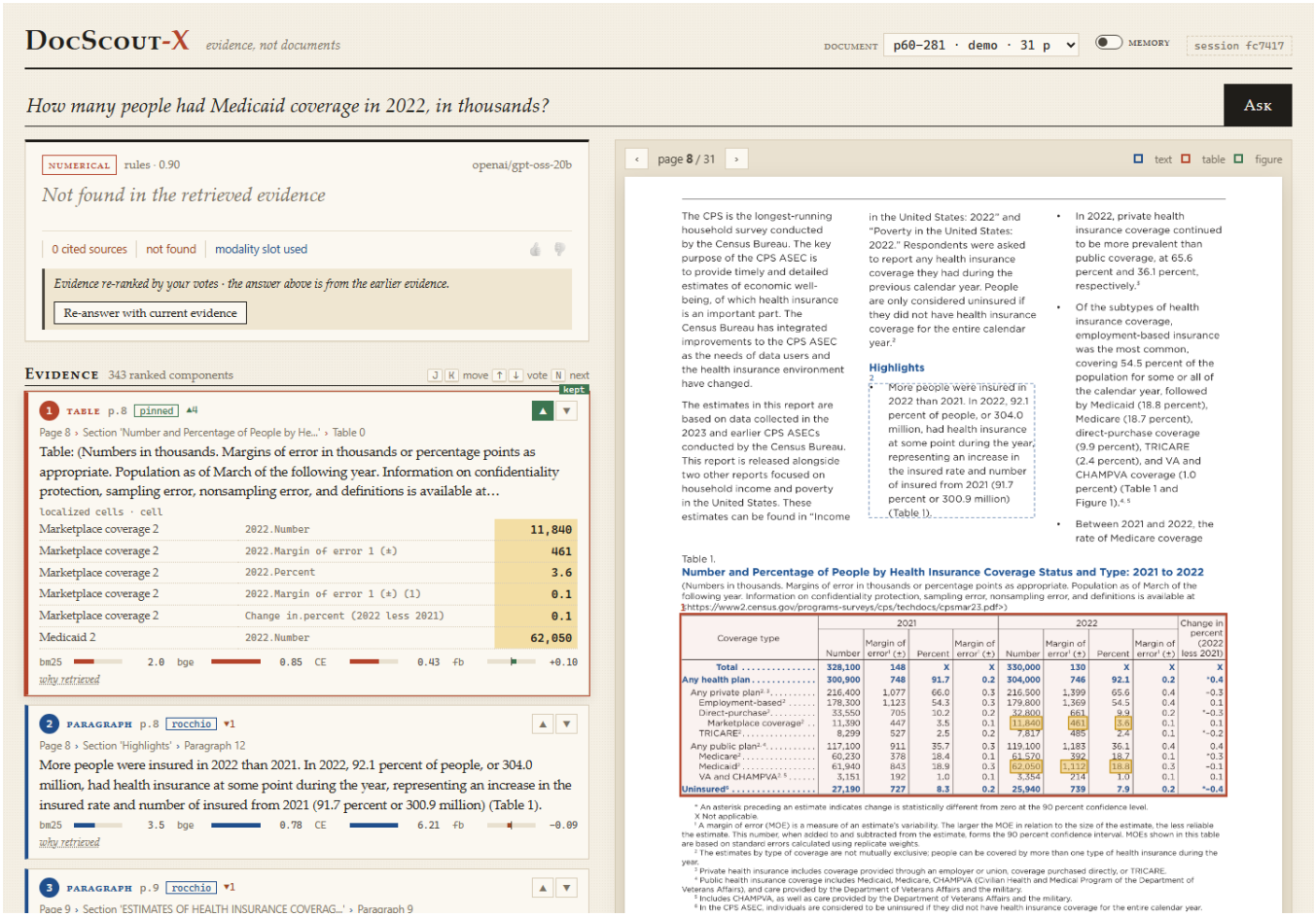


Figure 2. After down-voting the wrong table and up-voting Table 1: the table is pinned at #1 (moved up 4), other items show their rank change and the rocchio tag; the answer is marked as coming from the earlier evidence.

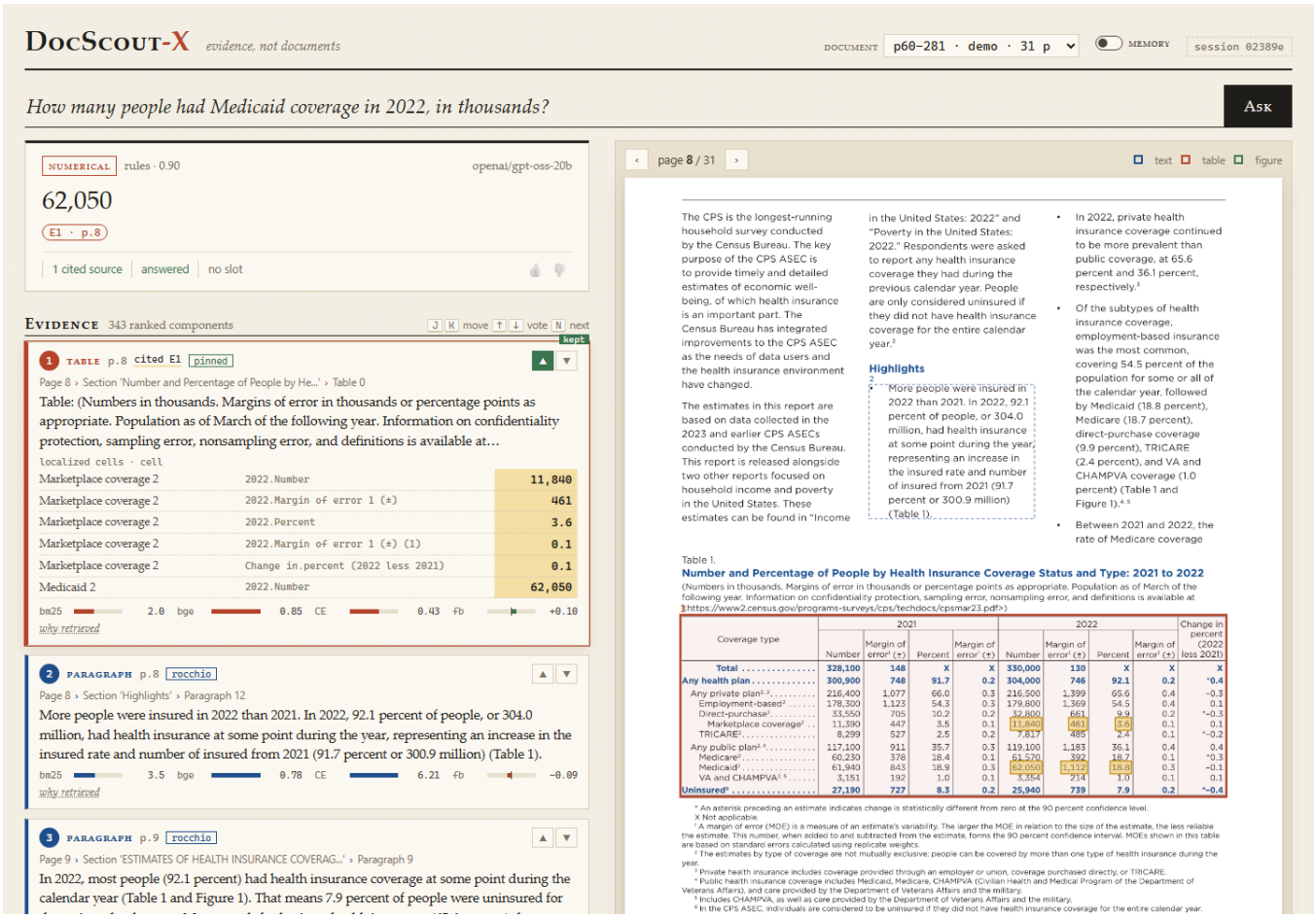


Figure 3. Re-answer in the same session: "62,050", citing E1 on page 8; the six located cells are highlighted on the page.

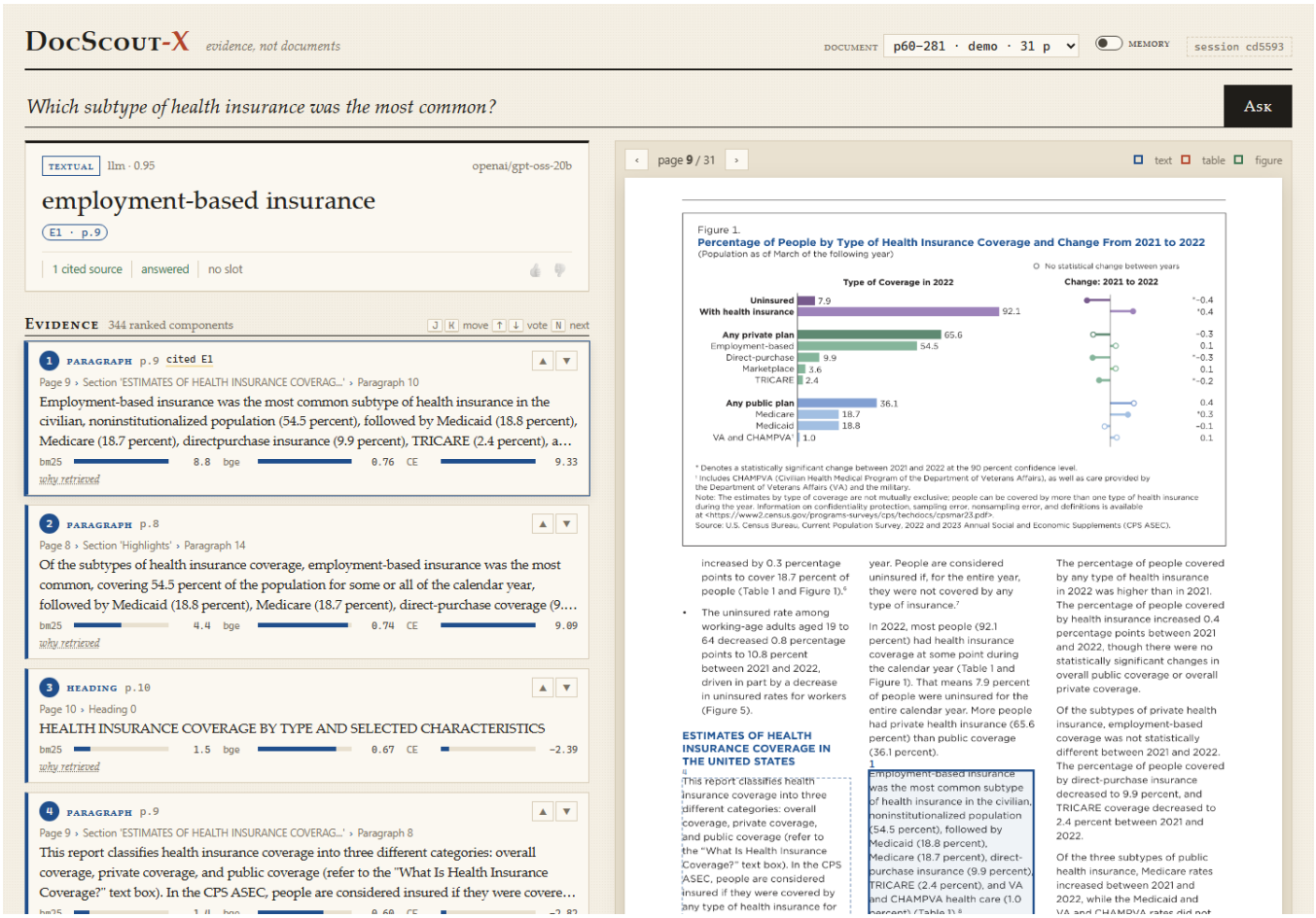


Figure 4. A textual question: "employment-based insurance", citing the paragraph on page 9.

Source: docs/reports/p7_evidence/ui_1_feedback_rerank.png, ui_2_reanswer.png, ui_3_textual.png (headless Edge, 1440x1000, real API)

10. Testing

Unit and integration tests run on CPU without downloading large models: the LLM, the cross-encoder and the retrievers are faked where needed, and the API is tested with FastAPI's TestClient on a fake pipeline. The UI was checked with a TypeScript build (tsc + vite build), real HTTP calls and headless-browser screenshots.

Test file	Tests	What it covers
tests/test_schema.py	5	Component / Evidence round trips, ids
tests/test_config.py	6	config paths, stores, keys
tests/test_download_utils.py	3	download helpers
tests/test_ingestion.py	15	parser mapping, serializer, graph
tests/test_ingest_isolation.py	1	crashing worker is isolated
tests/test_indexing.py	8	BM25 / dense / CLIP indexes, staleness
tests/test_eval.py	11	qrels, metrics, page-from-component
tests/test_classifier.py	17	rules, LLM fallback
tests/test_llm_client.py	9	cache, retries, rate limit
tests/test_retrieval.py	13	normalization, fusion, RRF, rerank tail
tests/test_split.py	2	document-level split
tests/test_answerer.py	12	prompt cap, citations, not found, bootstrap
tests/test_graph_expander.py	5	context expansion
tests/test_pipeline_smoke.py	6	end to end with fakes, feedback hook
tests/test_slots.py	6	slot reservation variants
tests/test_table_locator.py	14	cell / row / column selection
tests/test_rocchio.py	9	Rocchio math, pin / exclude, lambda blend, identity
tests/test_memory.py	9	votes store, memory gates
tests/test_api.py	7	all endpoints (TestClient)
tests/test_cell_boxes.py	2	cell boxes on a synthetic PDF
Total	**160**	pytest -> 160 passed (P7_REPORT 5)

Source: tests/; pytest --collect-only; P7_REPORT 5

11. Limitations and future work

- **Rocchio gain not significant on TEST:** +0.023 residual MRR [-0.048, +0.091] on 68 questions; DEV (34 questions, 5 with an up-vote) was too small for a reliable choice. Real user feedback, not simulated qrels votes, is needed.
- **Feedback-aware slots:** slot reservation picks the best *fused* table/figure and ignores feedback, so a memory penalty cannot remove a reserved item (left as is by decision; session Rocchio is unaffected).
- **Follow-up questions:** each question is answered on its own; there is no conversational context or query rewriting.
- **General-knowledge fallback:** by design the LLM answers only from evidence and says "Not found" otherwise; there is no optional, clearly labelled fallback to general knowledge.
- **PDF upload:** new documents are added from the command line (scripts.ingest --store demo, build_indexes --store demo --rebuild, API restart); the UI has no upload.
- **Cross-encoder vs tables/figures:** the CE prefers prose; on the demo it pushes Figure 2 to about rank 25 and Table 1 to rank 11, so the visual demo question fails at k = 5.
- **Lone-table normalization:** min-max lifts a lone table to the maximum branch score (hurts DEV page MRR before rerank).
- **Locator:** 13 of 24 oracle misses are row-label answers; superlative/count questions need comparison, not lookup; two rules were TEST-informed (clean estimate v1 quoted).
- **Memory:** small by design (cap 0.15 moved the gold table from rank 12 to 8); 5 of 34 unrelated fired pairs got slightly worse.
- **Cell boxes** only on digital PDFs; **CLIP** separates chart types weakly; **Groq** free-tier rate and daily token limits slow full ANLS runs.
- **Scale:** evaluation covers 49 documents / 293 questions of MP-DocVQA val; the full 150-document ingest was not run.

Source: P5-P7 reports, sections 8

12. How to run from a fresh clone (Windows)

```
python -m venv .venv          (Python 3.10)
.venv/Scripts/python -m pip install torch torchvision
    --index-url https://download.pytorch.org/whl/cpu
.venv/Scripts/python -m pip install -r requirements.txt
.venv/Scripts/python -m pip install -e .
copy .env.example .env
    (set LLM_PROVIDER=groq, LLM_API_KEY, LLM_MODEL=openai/gpt-oss-20b)
.venv/Scripts/python -m scripts.check_env
# behind HTTPS-scanning antivirus only: $env:SSL_CERT_FILE = "C:\path\to\ca-bundle.pem"

# data and evaluation corpus
.venv/Scripts/python -m scripts.download_data --limit 150
.venv/Scripts/python -m eval.select_eval_docs --target-pages 250 --max-pages 20
.venv/Scripts/python -m scripts.ingest --source mpdocvqa
    --docs-file data/raw/mpdocvqa/eval_docs.txt          (repeat until 49/49)
.venv/Scripts/python -m scripts.build_indexes --rebuild
.venv/Scripts/python -m eval.build_qrels

# demo store (put the PDF in data/raw/pdfs/ first)
.venv/Scripts/python -m scripts.ingest --source pdf --store demo
.venv/Scripts/python -m scripts.build_indexes --store demo --rebuild

# evaluation (see README for all systems and options)
.venv/Scripts/python -m eval.run_eval
.venv/Scripts/python -m eval.eval_answers
.venv/Scripts/python -m eval.eval_feedback --split dev --grid

# one question, the API and the UI
.venv/Scripts/python -m scripts.ask --doc p60-281
    "Which subtype of health insurance was the most common?"
.venv/Scripts/python -m uvicorn api.main:app --host 127.0.0.1 --port 8000
npm --prefix ui install
npm --prefix ui run dev        (open http://localhost:5173)
.venv/Scripts/python -m pytest
```

Source: README.md; P1-P7 reports, section 3

13. Team contribution split

Three equal parts by ownership of the real files (line counts at commit time). Shared by all: README.md, configs/config.yaml, docs/PHASES.md, docs/PROGRESS.md, the phase reports and this final package. Also in docs/final/CONTRIBUTIONS.md.

Member 1: [Name 1] ([Roll 1]): Ingestion, parsing, indexing, data and qrels

Phases: P1 setup + schema, P2 parsing/serialization/graph, P3 indexes, P4 data subset, qrels, ingest isolation. **Files owned:** 33 files, 3,622 lines.

config.py, schema.py, env_check.py, dataset/mpdocvqa.py, dataset/explore.py, ingestion/parser.py, ingestion/runner.py, ingestion/serializer.py, ingestion/validate.py, ingestion/visualize.py, graph/builder.py, indexing/base.py, indexing/bm25_index.py, indexing/dense_index.py, indexing/visual_index.py, indexing/build_all.py, indexing/search.py, scripts/check_env.py, scripts/download_data.py, scripts/explore_data.py, scripts/ingest.py, scripts/check_components.py, scripts/visualize_components.py, scripts/build_indexes.py, scripts/search_cli.py, eval/build_qrels.py, eval/select_eval_docs.py, tests/test_schema.py, tests/test_config.py, tests/test_download_utils.py, tests/test_ingestion.py, tests/test_ingest_isolation.py, tests/test_indexing.py

Member 2: [Name 2] ([Roll 2]): Retrieval, fusion, rerank, slots, table locator, graph context, evaluation

Phases: P4 metrics + baselines, P5 classifier/fusion/rerank/tuning/ablation, P6 locator, graph context, slots, ANLS + locator eval, P7 feedback eval. **Files owned:** 24 files, 3,570 lines.

retrieval/retrievers.py, retrieval/normalize.py, retrieval/fusion.py, retrieval/slots.py, rerank/cross_encoder.py, query/classifier.py, localization/table_locator.py, context/graph_expander.py, eval/metrics.py, eval/run_eval.py, eval/split.py, eval/tune_fusion.py, eval/eval_classifier.py, eval/plot_results.py, eval/eval_locator.py, eval/eval_answers.py, eval/eval_feedback.py, tests/test_retrieval.py, tests/test_classifier.py, tests/test_slots.py, tests/test_table_locator.py, tests/test_graph_expander.py, tests/test_eval.py, tests/test_split.py

Member 3: [Name 3] ([Roll 3]): LLM answering and grounding, feedback, API, React UI, tests

Phases: P5 LLM client, P6 prompts/answerer/provenance/pipeline/ask.py, P7 Rocchio, memory, cell boxes, API, UI. **Files owned:** 30 files, 3,527 lines.

reasoning/llm_client.py, reasoning/prompts.py, reasoning/answerer.py, pipeline.py, evidence/provenance.py, evidence/cell_boxes.py, feedback/session_store.py, feedback/rocchio.py, feedback/memory.py, api/main.py, ui/src/App.tsx, ui/src/api.ts, ui/src/styles.css, ui/src/components/AnswerPanel.tsx, ui/src/components/EvidenceCard.tsx, ui/src/components/PageViewer.tsx, ui/src/components/ScoreBars.tsx, ui/src/main.tsx, ui/index.html, ui/package.json, ui/tsconfig.json, ui/vite.config.ts, scripts/ask.py, tests/test_llm_client.py, tests/test_answerer.py, tests/test_pipeline_smoke.py, tests/test_rocchio.py, tests/test_memory.py, tests/test_api.py, tests/test_cell_boxes.py